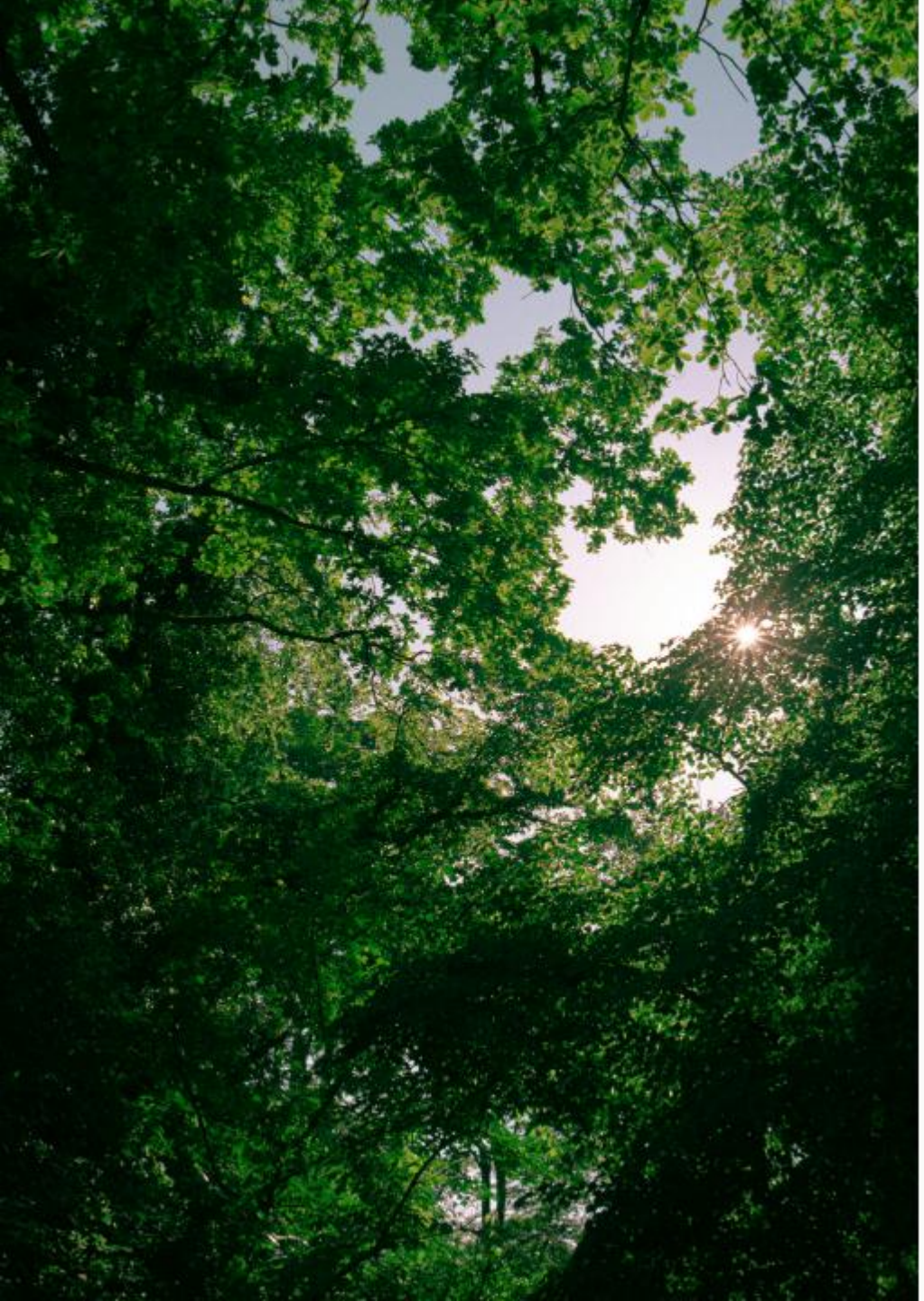
GRADE 11 GENERAL SCIENCE

The Chemistry of Life & Earth

Decoding the chemical equations behind biological survival, cellular metabolism, and environmental change.



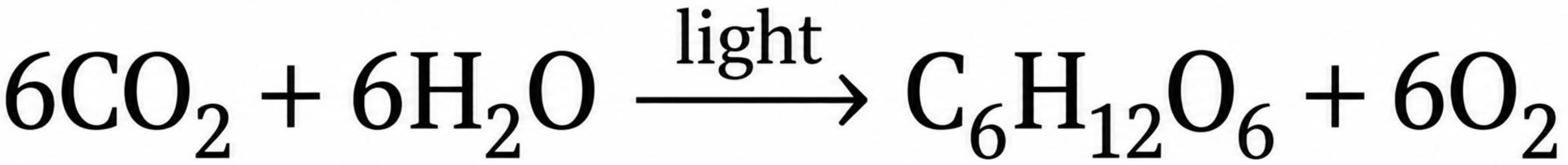
GLOBAL MECHANISMS

Earth is a Giant Chemical Reactor

Biological synthesis: Constructing organic matter from light.

Metabolic release: Breaking down molecules for cellular energy.

Environmental cycling: Moving elements across atmosphere and oceans.


6CO2 + 6H2O
light
C6H12O6 + 6O2

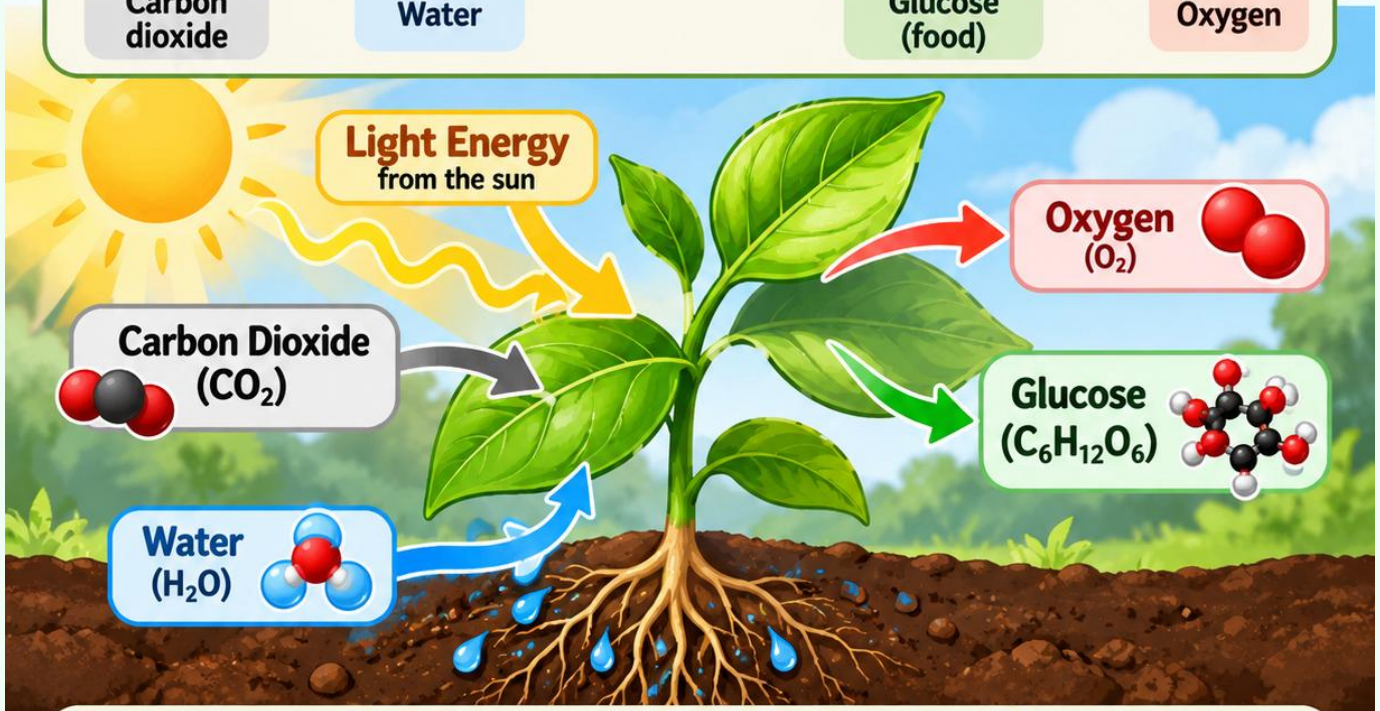
Photosynthesis

Photosynthesis is the process by which green plants, algae, and some bacteria use **light energy** from the sun to convert carbon dioxide and water into **glucose** (food), releasing **oxygen** as a byproduct.

Balanced Equation

$$6CO_2 + 6H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + 6O_2$$

$6CO_2$	$6H_2O$		$C_6H_{12}O_6$	$6O_2$
Carbon dioxide	Water		Glucose (food)	Oxygen



The plant uses **light energy** to convert carbon dioxide and water into **glucose (food)** and releases **oxygen**.

Deconstructing the Reaction



Reactants

Carbon Dioxide: Sourced from atmosphere.

Water: Absorbed through roots.

Products

Glucose: Stored chemical energy.

Oxygen: Released as metabolic byproduct.

BIOLOGICAL PROCESS 02

Cellular Respiration

The precise chemical reversal of photosynthesis.

Energy extraction: Breaks organic bonds.

ATP generation: Powers muscular and cellular work.

Continuous loop: Occurs in all living organisms.



$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$$

The Perfect Biological Loop



Photosynthesis

Stores Energy.

Consumes CO_2, releases O_2.





Respiration

Releases Energy.

Consumes O_2, releases CO_2.

CELLULAR REACTIONS

The Engine of Life: Metabolism

The sum of all chemical reactions occurring within our body cells to maintain life.

Catabolism

Breaking down. Destroys complex molecules to release energy (e.g., cellular respiration).

Anabolism

Building up. Uses energy to construct complex molecules (e.g., protein synthesis).



The Significance of Metabolism



Energy Production

Converts food into usable ATP to fuel all muscular and cellular work.



Growth & Repair

Synthesizes proteins and lipids required for cellular division and tissue healing.



Waste Elimination

Processes and flushes toxic metabolic byproducts from the cellular environment.



Thermal Regulation

Generates internal body heat through continuous chemical combustion.



Environmental Impacts

When biological and industrial equations fall out of equilibrium.



EXCESS CARBON

Ocean Acidification

$$CO_2 + H_2O \rightarrow H_2CO_3$$

Carbon Dioxide + Water → Carbonic Acid

The Impact: Elevated atmospheric CO_2 dissolves into oceans, lowering pH.

Consequence: Dissolves calcium carbonate structures in coral reefs.

INDUSTRIAL EMISSIONS

Formation of Acid Rain

$$SO_3 + H_2O \rightarrow H_2SO_4$$

Sulfur Trioxide + Water → Sulfuric Acid

The Impact: Fossil fuel combustion releases sulfur and nitrogen oxides into the air.

Consequence: Acidifies soil and lakes, destroying flora.



Ozone Layer Depletion

100,000

Ozone molecules destroyed by a single Chlorine atom.

$Cl + O_3 \rightarrow ClO + O_2$

CFCs release free chlorine, stripping an oxygen atom from protective ozone.

Student Activity

Chemical Detective: Match the equation to the global phenomenon.

Refer to the activity guidelines in SHS_GS_Q2_LE2.pdf



Analyze

Identify reactants and products.



Connect

Link to biological or environmental events.



Discuss

Debate impacts with your group.

Match the Process

Identify which phenomenon matches each chemical equation.

Chemical Equation / Action	Identify the Process
$CO_2 + H_2O \rightarrow H_2CO_3$	[Write answer here]
$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$	[Write answer here]
$Cl + O_3 \rightarrow ClO + O_2$	[Write answer here]
Converting food to usable ATP	[Write answer here]



Questions & Discussion

Chemistry is not just in a lab—it is the language of our living planet.

Image Sources



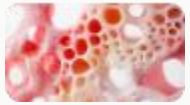
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